Crowd Synchrony and the "Wobbly" Millennium Bridge: A Replication Study of the Strogatz Model

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Abstract

On 10 June 2000, around 90,000 people decided to go on a walk and visit London's new Millennium Bridge. Within 3 days the thing turned from a celebration to an engineering case study: the deck began swaying side by side, and the bridge had to be closed. It eventually reopened in February 2002— so more "millennium +2 bridge" rather than just "millennium"—after engineers installed dampers. At the time, people thought the design of the bridge was the problem, until engineers realized it was a problem with synchronization: once the bridge starts moving laterally, pedestrians subconsciously adjust their gait for balance, and that adjustment feeds energy back into the bridge's side-to-side mode, amplifying the sway. A particularly interesting scientific explanation came from mathematician Steven Strogatz, who framed the problem as a single stability mechanism in which bridge wobbling and crowd synchrony emerge together to be explained by the mathematics of collective synchronization. The aim of this report is to replicate in Python the experiment made by Strogatz, and providing an explanation on its findings.

1 Why the Millennium Bridge a damped oscillator

A bridge is not a rigid object. It stores elastic energy when it bends (stiffness), has inertia because it has mass, and loses energy because joints/supports/materials are not perfectly lossless (damping). For small displacements, the dominant restoring force is approximately proportional to displacement, so the relevant lateral vibration mode can be idealized as a (weakly) damped harmonic oscillator.

A real bridge has infinitely many degrees of freedom, but engineers decompose motion into vibration modes. Near one resonant frequency, one mode often dominates, so the full structure can be reduced to one modal coordinate $X(t)$: "how excited is the dominant lateral mode?" This reduction leads to

$$M\ddot{X}(t) + B\dot{X}(t) + KX(t) = (\text{external forcing}), \tag{1}$$

where M, B, K are the modal mass, damping, and stiffness.

2 What happened on opening day: the affair as an instability

A damped oscillator at rest is stable: perturb it, and damping removes energy so the motion dies out. A crowd changes the story because pedestrians can inject energy back into the lateral mode. If the injected power per cycle exceeds the dissipated power, the equilibrium $X = 0$ becomes unstable and the bridge amplitude grows from small perturbations.

In ODEs, we can just rewrite (1) as a first-order system in $(X, \dot{X})$. Without forcing, $(0, 0)$ is a stable equilibrium. With pedestrian–bridge feedback, the effective damping can be reduced (even becoming negative), causing the fixed point to lose stability. In mechanical systems this typically corresponds to the birth of sustained oscillations (a limit-cycle-like state in the $(X, \dot{X})$ plane). Strogatz *et al.* explicitly frame the onset as a bifurcation where wobbling and synchrony emerge together as dual aspects of one instability.

3 Kuramoto model

Kuramoto's model is the canonical framework for collective synchronization:

$$\dot{\theta}_i = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i), \qquad i = 1, \ldots, N, \tag{2}$$

with order parameter

$$Re^{i\Phi} = \frac{1}{N}\sum_{j=1}^{N} e^{i\theta_j}, \qquad R \in [0, 1], \tag{3}$$

where $R \approx 0$ means incoherence and $R = \mathcal{O}(1)$ means partial synchrony.

The Millennium Bridge setup matches the Kuramoto worldview unusually well: pedestrians have gait phases; natural cadence varies across people; and the bridge provides a shared, mean-field coupling channel (everyone feels the same platform motion).

4 The Strogatz bridge model

Strogatz *et al.* couple a bridge oscillator to pedestrian gait phases:

$$M\ddot{X} + B\dot{X} + KX = \sum_{i=1}^{N} G\sin\Theta_i(t), \tag{4}$$

$$\dot{\Theta}_i(t) = \Omega_i + C\,A(t)\,\sin\big(\Psi(t) - \Theta_i(t) + \alpha\big), \quad i = 1, \ldots, N. \tag{5}$$

Bridge amplitude $A(t)$ and phase $\Psi(t)$ are defined at the bridge frequency $\Omega = \sqrt{K/M}$ by

$$X(t) = A(t)\sin\Psi(t), \qquad \dot{X}(t) = A(t)\,\Omega\cos\Psi(t), \tag{6}$$

equivalently $A(t) = \sqrt{X^2 + (\dot{X}/\Omega)^2}$.

Parameter meanings:

- M, B, K: modal mass/damping/stiffness; $\Omega = \sqrt{K/M}$ is the modal frequency.
- G: peak lateral forcing per pedestrian; N: number of pedestrians.
- Ω_i: natural stepping frequency of pedestrian i, drawn from density $P(\Omega)$.
- C: gait sensitivity to bridge motion amplitude; α: phase lag.

Instability mechanism in one sentence: small wobble ⇒ phase pulling ⇒ more coherence ⇒ larger net forcing ⇒ larger wobble.

5 Non-dimensionalizing the system

The supplementary analysis non-dimensionalizes the system to expose the controlling parameter groups and (crucially) a small parameter multiplying both coupling channels. This is what makes averaging methods legitimate and yields a tractable reduced description of onset.

6 Averaging, reduced slow dynamics, and the bifurcation

Under weak damping and near-resonant walking, averaging yields slow equations for bridge amplitude/phase and pedestrian relative phases. In the special case $\alpha = \pi/2$ and a symmetric unimodal frequency density, the steady-state conditions become isomorphic to Kuramoto's steady-state analysis. Two stationary regimes exist:

- motionless/desynchronized: $A = 0$, phases essentially uniform;
- wobbling/partially synchronized: $A > 0$, near-resonant pedestrians phase-lock.

The wobbling branch bifurcates from the motionless one; for bell-shaped distributions this onset is supercritical (smooth emergence of wobble amplitude).

7 Critical crowd size N_c

For the "simple case" $\alpha = \pi/2$ and symmetric $P(\Omega)$ about the bridge frequency, the predicted critical crowd size is

$$N_c = \frac{2B\Omega}{\pi\, G\, C\, P(\Omega)}. \tag{7}$$

Interpretation: larger damping B raises the threshold; larger forcing G, sensitivity C, or density near resonance $P(\Omega)$ lowers it.

8 How parameters move the bifurcation (what actually matters)

From (7):

- Bridge: increasing B increases N_c; changing M, K shifts Ω and therefore $P(\Omega)$.
- Crowd/coupling: increasing G or C lowers N_c.
- Frequency spread: broader cadence distributions generally make coherence harder, pushing thresholds up.

9 The instability feedback

The crowd's correlated excitation grows with bridge motion (often summarized as a velocity-proportional correlated force). In the Strogatz framework, increased synchrony yields a larger coherent forcing component aligned to pump energy efficiently into the bridge, which works as if it was an "effective negative damping" contribution.

10 Assumptions

Main idealizations:

- Single lateral mode (dominant degree of freedom, which don't let us think about vertical motion).
- Pedestrians modeled by phases only.
- Hypothetical but testable coupling law in (5).
- Weak damping / near resonance to justify averaging.

11 Numerical replication

11.1 Ramp experiment: $N(t)$, $A(t)$, and $R(t)$

The core diagnostic plot is Figure 1. The notebook reports:

$$N_c \approx 126.00, \qquad t_c \approx 66.10\,\text{s}.$$

Before t_c, $A(t)$ remains near zero and $R(t)$ stays small (incoherent fluctuations). After t_c, both $A(t)$ and $R(t)$ rise together, consistent with the predicted feedback-driven onset.

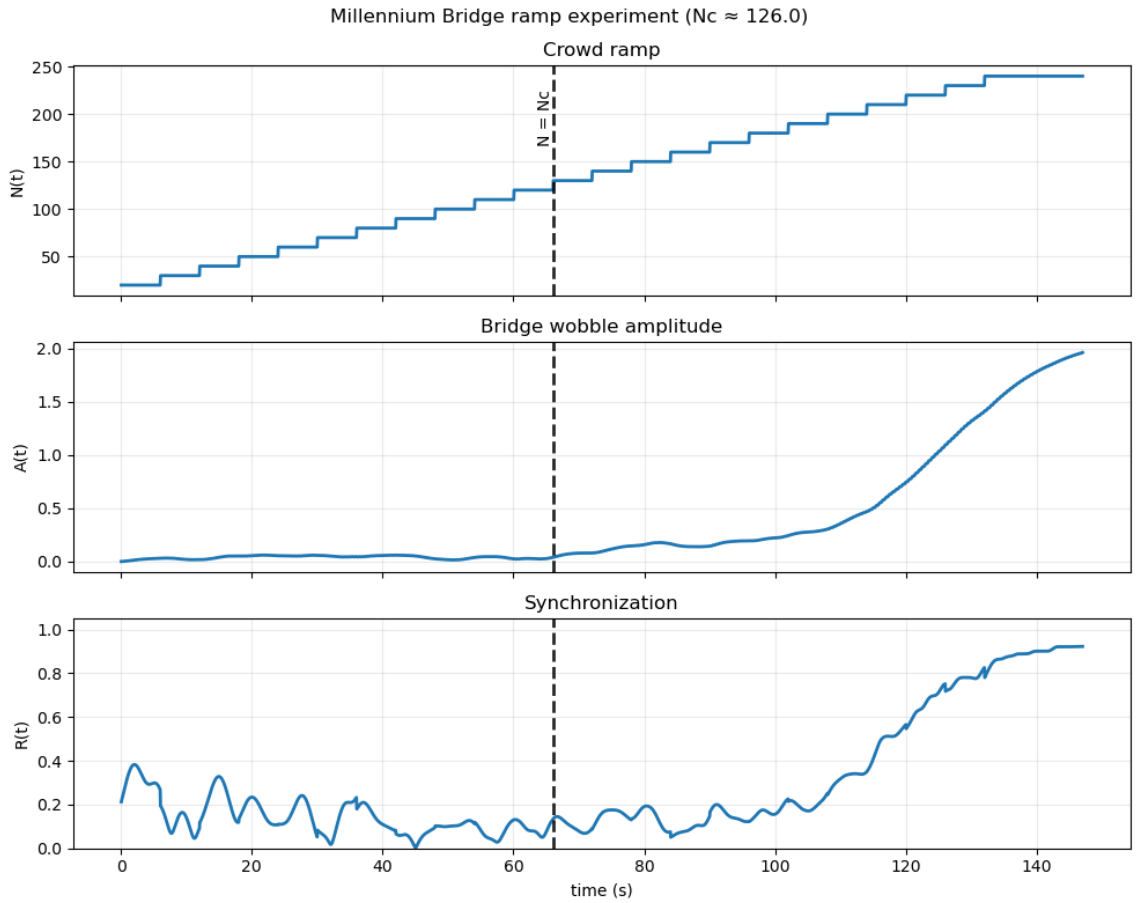


Figure 1: Crowd ramp $N(t)$ (top), wobble amplitude $A(t)$ (middle), and synchrony $R(t)$ (bottom). The vertical line marks the moment $t = t_c$ where $N(t)$ first reaches the theoretical N_c.

11.2 Onset vs crowd size (tail-step averaging)

Figure 2 uses tail averages per step to show onset more directly: mean tail A and mean tail R jump sharply near N_c.

11.3 Coupling sweep: effect of C

Equation (7) predicts $N_c \propto 1/C$. Figure 3 shows the notebook's sweep over three C values; larger C shifts onset leftward (smaller threshold crowd size).

11.4 Bifurcation-style plot: steady amplitude vs N

Figure 4 is the requested bifurcation-style visualization: steady wobble measure plotted across 100 N values around threshold, with the theoretical marker. The cloud of points stays near

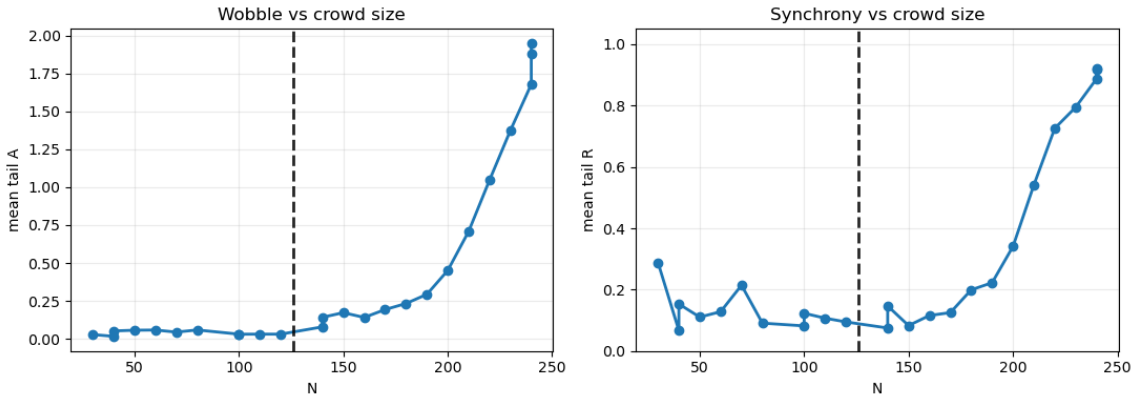


Figure 2: Mean tail amplitude and mean tail synchrony per crowd step vs N, with the dashed line marking N_c.

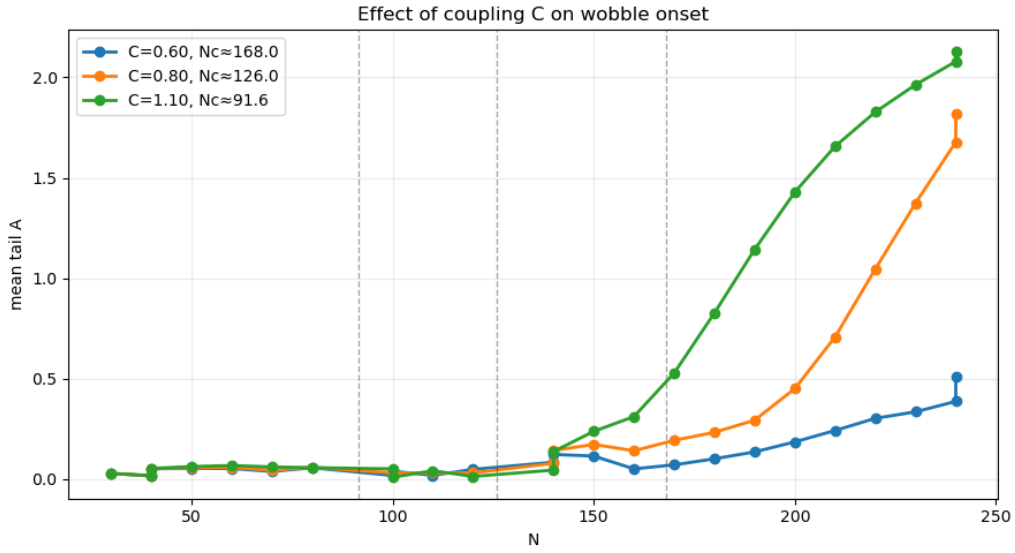


Figure 3: Effect of coupling sensitivity C on wobble onset. Each curve shows mean tail amplitude vs N, with dashed lines marking each run's theoretical N_c (as labeled in the plot legend).

zero below threshold and becomes clearly nonzero above it, with scatter due to randomness in phase/frequency realizations.

11.5 Numerical integral for $P(\Omega)$ and critical coupling C_{crit}

The notebook verifies $P(\Omega)$ via numerical integration, reporting:

$$P(\Omega)_\text{numeric} \approx 0.66490349, \quad P(\Omega)_\text{exact} \approx 0.66490380, \quad \text{relative error} \approx -4.63 \times 10^{-7}.$$

Rearranging (7) yields a critical coupling for fixed N:

$$C_{\text{crit}}(N) = \frac{2B\Omega}{\pi\, G\, N\, P(\Omega)}.$$

For the notebook's reference $N = 200$, it reports:

$$C_{\text{crit}} \approx 0.5040.$$

Finally, Figure 5 plots N_c as a function of C.

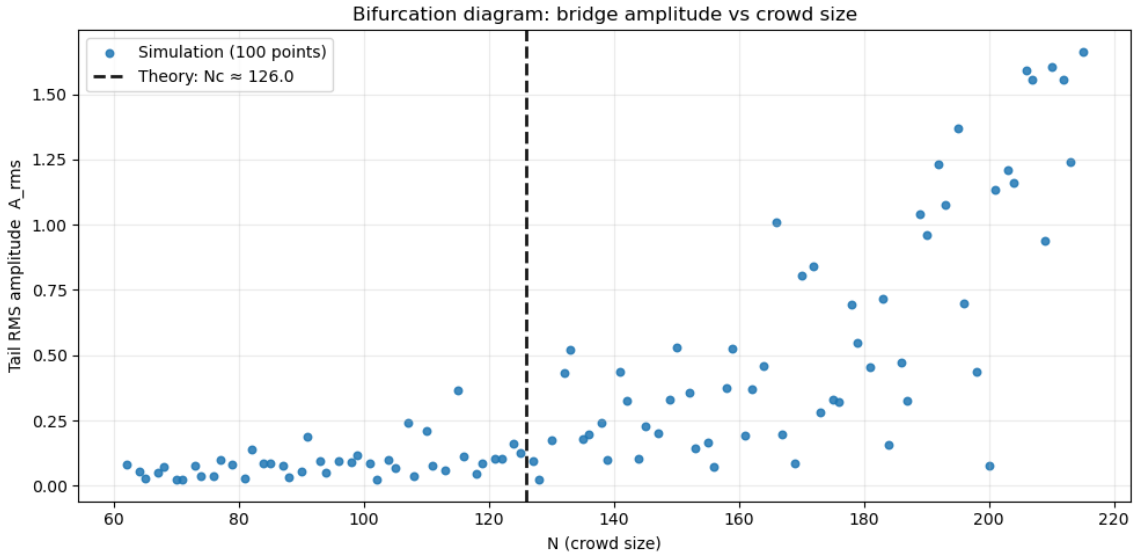


Figure 4: Empirical bifurcation-style plot of steady wobble amplitude vs N (100 values). The dashed marker indicates the theoretical threshold.

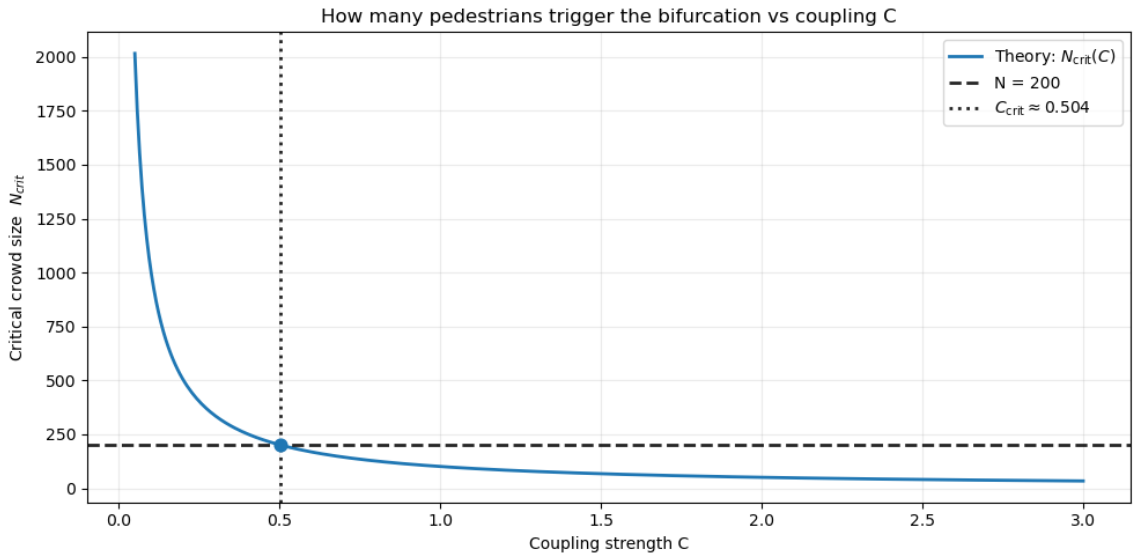


Figure 5: $N_c(C)$ computed by rearranging (7), with the reference point highlighted in the plot.

12 One clean ODE sentence (the whole story)

The Millennium Bridge can be modeled as a lightly damped lateral oscillator driven by many pedestrian phase oscillators. As crowd size (or effective coupling) passes a critical threshold, the motionless/desynchronized equilibrium loses stability and a stable wobbling state appears, consistent with a supercritical bifurcation in reduced slow dynamics and a Hopf-type "equilibrium $\rightarrow$ oscillation" transition in the physical $(X, \dot{X})$ dynamics.

References

[1] S. H. Strogatz, D. M. Abrams, A. McRobie, B. Eckhardt, and E. Ott, "Crowd synchrony on the Millennium Bridge," *Nature*, vol. 438, pp. 43–44, 2005.

[2] S. H. Strogatz *et al.*, Supplementary Information for "Crowd synchrony on the Millennium Bridge," 2005.

[3] I. Belykh *et al.*, "A new paradigm for emergent instability: the real story of the London Millennium Bridge," manuscript (submitted).